ModelIt!

Student Activity Pack

Can You Hear a Tsunami Coming?

How Early Detection Systems Save Lives from Giant Waves

Grade 04L2 | NGSS 4-PS4-3, 4-ESS3-2 | Cause and Effect

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **What do you know about tsunamis? What causes them and what happens when they reach shore?**

2. **How do you think scientists know a tsunami is coming before it arrives at the beach?**

3. **Draw what you think happens in the ocean between an earthquake and a tsunami hitting the coast.**

4. **Why do you think some coastal towns survive tsunamis with few deaths while others have many deaths?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Wave Energy | Distance from Shore | Detection Time | Warning Speed | Evacuation Success

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Far Tsunami, High Energy

Wave Energy at 80%, Distance from Shore at 80%

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Close Tsunami, High Energy

Wave Energy at 80%, Distance from Shore at 20%

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Far Tsunami, Low Energy

Wave Energy at 20%, Distance from Shore at 80%

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Coastal Shape — Whether the coastline is a flat beach, a bay, or a harbor. Some shapes funnel wave energy and make tsunamis worse, while others spread the energy out

Population Density — How many people live in the coastal area. More people means more people need to evacuate and more chances for traffic jams and confusion

Evacuation Practice — How often the community practices tsunami drills. Communities that practice regularly evacuate faster and more successfully

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

If a massive earthquake creates a tsunami far out at sea, how do warning systems detect it in time to save people onshore? Use evidence from your simulation to support your answers.

1 Why is distance from shore such an important factor in tsunami survival?

2 If you cannot control how strong a tsunami is, what CAN you control to save more lives?

3 How is a tsunami warning system similar to a fire alarm in your school?

4 What would you tell a friend who lives near the ocean about tsunami safety?

5 How did your model help you understand why some tsunami events are more deadly than others?

8 STEM Challenge: Design a Tsunami Warning System

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YOUR ENGINEERING MISSION

Design a warning system that gives a coastal town the maximum amount of time to evacuate before tsunami waves arrive.

THE SCENARIO

A coastal town in the Pacific has experienced two tsunamis in the past 20 years. The first tsunami started 500 miles away and killed no one because there was plenty of warning time. The second started just 50 miles offshore and killed 200 people because the warning came too late. The town council has hired your engineering team to design a better warning system that can save lives even when tsunamis start close to shore. Consider detection speed, communication methods, and evacuation routes.

Engineering Design Thinking

1. WHERE WOULD YOU PLACE OCEAN SENSORS FOR THE EARLIEST POSSIBLE DETECTION OF NEARBY EARTHQUAKES?

2. WHAT COMMUNICATION METHODS WOULD YOU USE TO REACH EVERYONE QUICKLY (SIRENS, PHONE ALERTS, RADIO)?

3. HOW WOULD YOU DESIGN EVACUATION ROUTES SO THEY DO NOT GET JAMMED WITH PEOPLE ALL TRYING TO LEAVE AT ONCE?

4. WHAT COULD YOU BUILD IN THE TOWN (VERTICAL EVACUATION TOWERS, ELEVATED SHELTERS) FOR PEOPLE WHO CANNOT REACH HIGH GROUND IN TIME?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Emergency Management Specialists design disaster warning systems and evacuation plans that save lives. They work with scientists, engineers, and government agencies to prepare communities for natural disasters like tsunamis, earthquakes, and hurricanes. They earn $55,000-$95,000/year.